5.2 HBE-COMBO III Pin map

1) Altera Device Module

1. Clock				
Net Name	***PAD***	***From***	***To***	***비고***
CLK_MAIN[0]	N2	Oscillator	FPGA Device	MAIN CLOCK
CLK_MAIN[1]	N25	Oscillator	FPGA Device	MAIN CLOCK
CLK_USER[0]	A13	Oscillator	FPGA Device	USER CLOCK
CLK_USER[1]	D13	Oscillator	FPGA Device	USER CLOCK
2. RESET				
Net Name	***PAD***	***From***	***To***	***비고***
RESET	C5	Button Switch	FPA Device	System Switch
3. LED				
Net Name	***PAD***	***From***	***To***	***비고***
LED_D[0]	AE11	FPGA Module	Base board	Display LED
LED_D[1]	AE10	FPGA Module	Base board	Display LED
LED_D[2]	AF9	FPGA Module	Base board	Display LED
LED_D[3]	AF8	FPGA Module	Base board	Display LED
LED_D[4]	AF7	FPGA Module	Base board	Display LED
LED_D[5]	AF6	FPGA Module	Base board	Display LED
LED_D[6]	AF5	FPGA Module	Base board	Display LED
LED_D[7]	B3	FPGA Module	Base board	Display LED
LED_D[8]	B2	FPGA Module	Base board	Display LED
LED_D[9]	AF13	FPGA Module	Base board	Display LED
LED_D[10]	Y25	FPGA Module	Base board	Display LED
LED_D[11]	AF23	FPGA Module	Base board	Display LED
LED_D[12]	G1	FPGA Module	Base board	Display LED
LED_D[13]	G2	FPGA Module	Base board	Display LED
LED_D[14]	AA24	FPGA Module	Base board	Display LED
LED_D[15]	AF10	FPGA Module	Base board	Display LED
4. Segment(Array)				
Net Name	***PAD***	***From***	***To***	***비고***
SEG_SCAN[0]	T3	FPGA Module	Base board	Display Segment
SEG_SCAN[1]	T2	FPGA Module	Base board	Display Segment
SEG_SCAN[2]	R7	FPGA Module	Base board	Display Segment
SEG_SCAN[3]	R6	FPGA Module	Base board	Display Segment
SEG_SCAN[4]	U3	FPGA Module	Base board	Display Segment

SEG_SCAN[5]	U2	FPGA Module	Base board	Display Segment
SEG_SCAN[6]	U1	FPGA Module	Base board	Display Segment
SEG_SCAN[7]	T7	FPGA Module	Base board	Display Segment
SEG_DATA_A	M5	FPGA Module	Base board	Display Segment
SEG_DATA_B	M4	FPGA Module	Base board	Display Segment
SEG_DATA_C	R5	FPGA Module	Base board	Display Segment
SEG_DATA_D	R4	FPGA Module	Base board	Display Segment
SEG_DATA_E	R3	FPGA Module	Base board	Display Segment
SEG_DATA_F	R2	FPGA Module	Base board	Display Segment
SEG_DATA_G	P7	FPGA Module	Base board	Display Segment
SEG_DATA_DP	T4	FPGA Module	Base board	Display Segment
5. Segment				
Net Name	***PAD***	***From***	***To***	***비고***
SEG_DATA_A	M3	FPGA Module	Base board	Display Segment
SEG_DATA_B	M2	FPGA Module	Base board	Display Segment
SEG_DATA_C	L7	FPGA Module	Base board	Display Segment
SEG_DATA_D	L6	FPGA Module	Base board	Display Segment
SEG_DATA_E	L4	FPGA Module	Base board	Display Segment
SEG_DATA_F	P6	FPGA Module	Base board	Display Segment
SEG_DATA_G	P4	FPGA Module	Base board	Display Segment
SEG_DATA_DP	P3	FPGA Module	Base board	Display Segment
6. Vacuum Fluorescent Display(VFD)				
Net Name	***PAD***	***From***	***To***	***비고***
VFD_D[0]	K4	FPGA Module	Base board	Display LCD
VFD_D[1]	K3	FPGA Module	Base board	Display LCD
VFD_D[2]	K2	FPGA Module	Base board	Display LCD
VFD_D[3]	K1	FPGA Module	Base board	Display LCD
VFD_D[4]	L3	FPGA Module	Base board	Display LCD
VFD_D[5]	L2	FPGA Module	Base board	Display LCD
VFD_D[6]	K8	FPGA Module	Base board	Display LCD
VFD_D[7]	K7	FPGA Module	Base board	Display LCD
VFD_E	K5	FPGA Module	Base board	Display LCD
VFD_RW	J4	FPGA Module	Base board	Display LCD
VFD_RS	J5	FPGA Module	Base board	Display LCD
7. DOT				
Net Name	***PAD***	***From***	***To***	***비고***
DOT_D[0]	L23	FPGA Module	Base board	Display Dot-matrix

DOT_D[1]	C21	FPGA Module	Base board	Display Dot-matrix
DOT_D[2]	C19	FPGA Module	Base board	Display Dot-matrix
DOT_D[3]	L21	FPGA Module	Base board	Display Dot-matrix
DOT_D[4]	L20	FPGA Module	Base board	Display Dot-matrix
DOT_D[5]	C17	FPGA Module	Base board	Display Dot-matrix
DOT_D[6]	D18	FPGA Module	Base board	Display Dot-matrix
DOT_D[7]	L19	FPGA Module	Base board	Display Dot-matrix
DOT_D[8]	K26	FPGA Module	Base board	Display Dot-matrix
DOT_D[9]	D17	FPGA Module	Base board	Display Dot-matrix
DOT_D[10]	D16	FPGA Module	Base board	Display Dot-matrix
DOT_D[11]	M21	FPGA Module	Base board	Display Dot-matrix
DOT_D[12]	M20	FPGA Module	Base board	Display Dot-matrix
DOT_D[13]	D15	FPGA Module	Base board	Display Dot-matrix
DOT_COM[0]	B23	FPGA Module	Base board	Display Dot-matrix
DOT_COM[1]	K25	FPGA Module	Base board	Display Dot-matrix
DOT_COM[2]	K24	FPGA Module	Base board	Display Dot-matrix
DOT_COM[3]	B22	FPGA Module	Base board	Display Dot-matrix
DOT_COM[4]	B21	FPGA Module	Base board	Display Dot-matrix
DOT_COM[5]	K23	FPGA Module	Base board	Display Dot-matrix
DOT_COM[6]	K22	FPGA Module	Base board	Display Dot-matrix
DOT_COM[7]	C23	FPGA Module	Base board	Display Dot-matrix
DOT_COM[8]	C22	FPGA Module	Base board	Display Dot-matrix
DOT_COM[9]	K21	FPGA Module	Base board	Display Dot-matrix
8. PIEZO				
Net Name	***PAD***	***From***	***To***	***비고***
PIEZO	K6	FPGA Module	Base board	Speaker control
9. SRAM				
Net Name	***PAD***	***From***	***To***	***비고***
SRAM_A[0]	B24	FPGA Module	Base board	Memory control
SRAM_A[1]	B25	FPGA Module	Base board	Memory control
SRAM_A[2]	C24	FPGA Module	Base board	Memory control
SRAM_A[3]	C25	FPGA Module	Base board	Memory control
SRAM_A[4]	D23	FPGA Module	Base board	Memory control
SRAM_A[5]	D25	FPGA Module	Base board	Memory control
SRAM_A[6]	D26	FPGA Module	Base board	Memory control
SRAM_A[7]	E22	FPGA Module	Base board	Memory control
SRAM_A[8]	E23	FPGA Module	Base board	Memory control
SRAM_A[9]	E24	FPGA Module	Base board	Memory control

SRAM_A[10]	E25	FPGA Module	Base board	Memory control
SRAM_A[11]	E26	FPGA Module	Base board	Memory control
SRAM_A[12]	F20	FPGA Module	Base board	Memory control
SRAM_A[13]	F24	FPGA Module	Base board	Memory control
SRAM_A[14]	F25	FPGA Module	Base board	Memory control
SRAM_A[15]	F26	FPGA Module	Base board	Memory control
SRAM_A[16]	G21	FPGA Module	Base board	Memory control
SRAM_A[17]	G22	FPGA Module	Base board	Memory control
SRAM_D[0]	P23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[1]	P24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[2]	R20	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[3]	R24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[4]	R25	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[5]	T20	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[6]	T21	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[7]	T22	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[8]	T23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[9]	T24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[10]	T25	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[11]	U20	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[12]	U21	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[13]	U22	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[14]	U23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[15]	U24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[16]	U25	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[17]	U26	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[18]	V22	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[19]	V23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[20]	V24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[21]	V25	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[22]	V26	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[23]	W21	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[24]	W23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[25]	W24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[26]	W25	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[27]	W26	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[28]	Y21	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[29]	Y22	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[30]	Y23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[31]	Y24	FPGA & Base board	FPGA & Base board	Memory control

SRAM1_LB	G23	FPGA Module	Base board	Memory control
SRAM1_UB	G24	FPGA Module	Base board	Memory control
SRMA2_LB	H23	FPGA Module	Base board	Memory control
SRAM2_UB	H24	FPGA Module	Base board	Memory control
SRAM_OE	J21	FPGA Module	Base board	Memory control
SRAM_WE	J20	FPGA Module	Base board	Memory control
SRAM_CS	H26	FPGA Module	Base board	Memory control
10. Flash Memory				
Net Name	***PAD***	***From***	***To***	***비고***
Flash_A[1]	B24	FPGA Module	Base board	Memory control
Flash_A[2]	B25	FPGA Module	Base board	Memory control
Flash_A[3]	C24	FPGA Module	Base board	Memory control
Flash_A[4]	C25	FPGA Module	Base board	Memory control
Flash_A[5]	D23	FPGA Module	Base board	Memory control
Flash_A[6]	D25	FPGA Module	Base board	Memory control
Flash_A[7]	D26	FPGA Module	Base board	Memory control
Flash_A[8]	E22	FPGA Module	Base board	Memory control
Flash_A[9]	E23	FPGA Module	Base board	Memory control
Flash_A[10]	E24	FPGA Module	Base board	Memory control
Flash_A[11]	E25	FPGA Module	Base board	Memory control
Flash_A[12]	E26	FPGA Module	Base board	Memory control
Flash_A[13]	F20	FPGA Module	Base board	Memory control
Flash_A[14]	F24	FPGA Module	Base board	Memory control
Flash_A[15]	F25	FPGA Module	Base board	Memory control
Flash_A[16]	F26	FPGA Module	Base board	Memory control
Flash_A[17]	G21	FPGA Module	Base board	Memory control
Flash_A[18]	G22	FPGA Module	Base board	Memory control
Flash_A[19]	G23	FPGA Module	Base board	Memory control
Flash_A[20]	G24	FPGA Module	Base board	Memory control
Flash_DQ[0]	P23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[1]	P24	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[2]	R20	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[3]	R24	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[4]	R25	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[5]	T20	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[6]	T21	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[7]	T22	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[8]	T23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[9]	T24	FPGA & Base board	FPGA & Base board	Memory control

Flash_DQ[10]	T25	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[11]	U20	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[12]	U21	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[13]	U22	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[14]	U23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[15]	U24	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[16]	U25	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[17]	U26	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[18]	V22	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[19]	V23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[20]	V24	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[21]	V25	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[22]	V26	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[23]	W21	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[24]	W23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[25]	W24	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[26]	W25	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[27]	W26	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[28]	Y21	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[29]	Y22	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[30]	Y23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[31]	Y24	FPGA & Base board	FPGA & Base board	Memory control
/Flash_OE	G25	FPGA Module	Base board	Memory control
/Flash_WE	G26	FPGA Module	Base board	Memory control
/Flash_CS	H21	FPGA Module	Base board	Memory control
/Flash_RP	H23	FPGA Module	Base board	Memory control
/Flash_RB	H24	FPGA Module	Base board	Memory control
/Flash_WP	H25	FPGA Module	Base board	Memory control
11. Step Motor				
Net Name	***PAD***	***From***	***To***	***비고***
STEP_DATA[0]	AA23	FPGA Module	Base board	Motor control (A)
STEP_DATA[1]	AB24	FPGA Module	Base board	Motor control (B)
STEP_DATA[2]	Y26	FPGA Module	Base board	Motor control (/A)
STEP_DATA[3]	AB23	FPGA Module	Base board	Motor control (/B)
MOTOR_SENSOR	AA26	Base board	FPGA Module	Magnetic sensor
12. SD Connector				
Net Name	***PAD***	***From***	***To***	***비고***
SD_DATA[0]	J1	FPGA & Base board	FPGA & Base board	Memory control

SD_DATA[1]	H6	FPGA & Base board	FPGA & Base board	Memory control
SD_DATA[2]	H4	FPGA & Base board	FPGA & Base board	Memory control
SD_DATA[3]	J8	FPGA & Base board	FPGA & Base board	Memory control
SD_CMD	J3	FPGA & Base board	FPGA & Base board	Memory control
SD_CLK	J2	FPGA & Base board	FPGA & Base board	Memory control
13. DIP SWITCH				
Net Name	***PAD***	***From***	***To***	***비고***
DIP_D[0]	T6	Base board	FPGA Module	Input Equipment
DIP_D[1]	V1	Base board	FPGA Module	Input Equipment
DIP_D[2]	U7	Base board	FPGA Module	Input Equipment
DIP_D[3]	U6	Base board	FPGA Module	Input Equipment
DIP_D[4]	U5	Base board	FPGA Module	Input Equipment
DIP_D[5]	U4	Base board	FPGA Module	Input Equipment
DIP_D[6]	V6	Base board	FPGA Module	Input Equipment
DIP_D[7]	V5	Base board	FPGA Module	Input Equipment
14. BUTTON SWITCH				
Net Name	***PAD***	***From***	***To***	***비고***
BUTTON_SW[0]	V4	Base board	FPGA Module	Input Equipment
BUTTON_SW[1]	V3	Base board	FPGA Module	Input Equipment
BUTTON_SW[2]	V2	Base board	FPGA Module	Input Equipment
BUTTON_SW[3]	W4	Base board	FPGA Module	Input Equipment
BUTTON_SW[4]	W3	Base board	FPGA Module	Input Equipment
BUTTON_SW[5]	W2	Base board	FPGA Module	Input Equipment
BUTTON_SW[6]	W1	Base board	FPGA Module	Input Equipment
BUTTON_SW[7]	V7	Base board	FPGA Module	Input Equipment
BUTTON_SW[8]	Y4	Base board	FPGA Module	Input Equipment
BUTTON_SW[9]	Y3	Base board	FPGA Module	Input Equipment
BUTTON_SW[10]	Y2	Base board	FPGA Module	Input Equipment
BUTTON_SW[11]	Y1	Base board	FPGA Module	Input Equipment
BUTTON_SW[12]	W6	Base board	FPGA Module	Input Equipment
BUTTON_SW[13]	AA4	Base board	FPGA Module	Input Equipment
BUTTON_SW[14]	AA3	Base board	FPGA Module	Input Equipment
BUTTON_SW[15]	AA2	Base board	FPGA Module	Input Equipment
15. Humidity & Temp Sensor				
Net Name	***PAD***	***From***	***To***	***비고***
SENSOR_CLK	J7	FPGA Module	Base board	Serial clock input
SENSOR_DATA	J6	FPGA & Base board	FPGA & Base board	Serial data

16. TFT-LCD				
Net Name	***PAD***	***From***	***To***	***비고***
TFTLCD_D[0]	AC8	FPGA Module	TFTLCD Module	TFTLCD RED[0]
TFTLCD_D[1]	AC7	FPGA Module	TFTLCD Module	TFTLCD RED[1]
TFTLCD_D[2]	AB12	FPGA Module	TFTLCD Module	TFTLCD RED[2]
TFTLCD_D[3]	AB10	FPGA Module	TFTLCD Module	TFTLCD RED[3]
TFTLCD_D[4]	AB8	FPGA Module	TFTLCD Module	TFTLCD RED[4]
TFTLCD_D[5]	AA12	FPGA Module	TFTLCD Module	TFTLCD RED[5]
TFTLCD_D[6]	AC10	FPGA Module	TFTLCD Module	TFTLCD RED[6]
TFTLCD_D[7]	AC9	FPGA Module	TFTLCD Module	TFTLCD RED[7]
TFTLCD_D[8]	AC6	FPGA Module	TFTLCD Module	TFTLCD GREEN[0]
TFTLCD_D[9]	AD6	FPGA Module	TFTLCD Module	TFTLCD GREEN[1]
TFTLCD_D[10]	AD5	FPGA Module	TFTLCD Module	TFTLCD GREEN[2]
TFTLCD_D[11]	AD4	FPGA Module	TFTLCD Module	TFTLCD GREEN[3]
TFTLCD_D[12]	AC12	FPGA Module	TFTLCD Module	TFTLCD GREEN[4]
TFTLCD_D[13]	AC11	FPGA Module	TFTLCD Module	TFTLCD GREEN[5]
TFTLCD_D[14]	AD12	FPGA Module	TFTLCD Module	TFTLCD GREEN[6]
TFTLCD_D[15]	AD11	FPGA Module	TFTLCD Module	TFTLCD GREEN[7]
TFTLCD_D[16]	AD10	FPGA Module	TFTLCD Module	TFTLCD_BLUE[0]
TFTLCD_D[17]	AD8	FPGA Module	TFTLCD Module	TFTLCD_BLUE[1]
TFTLCD_D[18]	AD7	FPGA Module	TFTLCD Module	TFTLCD_BLUE[2]
TFTLCD_D[19]	AE9	FPGA Module	TFTLCD Module	TFTLCD_BLUE[3]
TFTLCD_D[20]	AE8	FPGA Module	TFTLCD Module	TFTLCD_BLUE[4]
TFTLCD_D[21]	AE7	FPGA Module	TFTLCD Module	TFTLCD_BLUE[5]
TFTLCD_D[22]	AE6	FPGA Module	TFTLCD Module	TFTLCD_BLUE[6]
TFTLCD_D[23]	AE5	FPGA Module	TFTLCD Module	TFTLCD_BLUE[7]
TFTLCD_ENABLE	AF4	FPGA Module	TFTLCD Module	Enable signal
TFTLCD_LCLK	AE13	FPGA Module	TFTLCD Module	HSYNC
TFTLCD_FCLK	AE12	FPGA Module	TFTLCD Module	VSYNC
TFTLCD_PCLK	AA6	FPGA Module	TFTLCD Module	CLOCK
TFTLCD_RESET	AA9	FPGA Module	TFTLCD Module	RESET
17. Image Sensor				
Net Name	***PAD***	***From***	***To***	***비고***
CIS DATA[]	AC2	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	AC1	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	AB4	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	AB3	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	Y10	FPGA Module	TFTLCD Module	Image out Data

CIS DATA[]	AE3	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	AE2	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	AD3	FPGA Module	TFTLCD Module	Image out Data
CIS_SCK	AD2	FPGA Module	TFTLCD Module	Serial clock
CIS_SDA	AA11	FPGA Module	TFTLCD Module	Serial data
CIS_STROBE	AA10	FPGA Module	TFTLCD Module	Strobe signal
CIS_VCLK	Y12	FPGA Module	TFTLCD Module	Pixel clock
CIS_LVALID	Y11	FPGA Module	TFTLCD Module	Line valid
CIS_FSYNC	AC5	FPGA Module	TFTLCD Module	Frame valid
CIS_MCLK	AA7	FPGA Module	TFTLCD Module	Master clock
CIS_RESET	AA9	FPGA Module	TFTLCD Module	reset
18. TFT-LCD Touch				
Net Name	***PAD***	***From***	***To***	***비고***
TOUCH_DCLK	AC3	FPGA Module	TFTLCD Module	External clock
TOUCH_DIN	AA1	FPGA Module	TFTLCD Module	Data input
TOUCH_DOUT	AB1	FPGA Module	TFTLCD Module	Data output
TOUCH_BUSY	Y5	FPGA Module	TFTLCD Module	Busy output
TOUCH_IRQ	AA5	FPGA Module	TFTLCD Module	Pen interrupt
TOUCH_CS	AB2	FPGA Module	TFTLCD Module	Chip Select
19. OLED				
Net Name	***PAD***	***From***	***To***	***비고***
OLED_DATA[0]	L24	FPGA Module	OLED Module	OLED Data
OLED_DATA[1]	N20	FPGA Module	OLED Module	OLED Data
OLED_DATA[2]	D21	FPGA Module	OLED Module	OLED Data
OLED_DATA[3]	D20	FPGA Module	OLED Module	OLED Data
OLED_DATA[4]	M25	FPGA Module	OLED Module	OLED Data
OLED_DATA[5]	M24	FPGA Module	OLED Module	OLED Data
OLED_DATA[6]	D18	FPGA Module	OLED Module	OLED Data
OLED_DATA[7]	D17	FPGA Module	OLED Module	OLED Data
OLED_CS	D14	FPGA Module	OLED Module	Chip select
OLED_WN	E18	FPGA Module	OLED Module	Write enable
OLED_RN	E15	FPGA Module	OLED Module	Read enable
OLED_DC	L25	FPGA Module	OLED Module	Data/Command
OLED_RESET	M19	FPGA Module	OLED Module	Reset pin
20. Audio Codec				
Net Name	***PAD***	***From***	***To***	***비고***
AIC_CS	M23	FPGA Module	Audio Codec	Latch/Add Select

AIC_MODE	M22	FPGA Module	Audio Codec	Serial interface
AIC_ADI	F15	FPGA Module	Audio Codec	Serial Data input
AIC_SCLK	F14	FPGA Module	Audio Codec	Serial Data clock
AIC_MCLK	F21	FPGA Module	Audio Codec	External clock
AIC_LRCOUT	N23	FPGA Module	Audio Codec	ADC clock signal
AIC_DOUT	F13	FPGA Module	Audio Codec	Serial Data output
AIC_LRCIN	E20	FPGA Module	Audio Codec	DAC clock signal
AIC_DIN	N24	FPGA Module	Audio Codec	Serial Data input
AIC_BCLK	V21	FPGA Module	Audio Codec	Serial bit clock
21. VGA PORT				
Net Name	***PAD***	***From***	***To***	***비고***
VGA_B[0]	G17	FPGA Module	VGA Port	Display Equipment
VGA_B[1]	G16	FPGA Module	VGA Port	Display Equipment
VGA_B[2]	B20	FPGA Module	VGA Port	Display Equipment
VGA_B[3]	B19	FPGA Module	VGA Port	Display Equipment
VGA_B[4]	J22	FPGA Module	VGA Port	Display Equipment
VGA_B[5]	K19	FPGA Module	VGA Port	Display Equipment
VGA_B[6]	B18	FPGA Module	VGA Port	Display Equipment
VGA_B[7]	B17	FPGA Module	VGA Port	Display Equipment
VGA_R[0]	A20	FPGA Module	VGA Port	Display Equipment
VGA_R[1]	A19	FPGA Module	VGA Port	Display Equipment
VGA_R[2]	A18	FPGA Module	VGA Port	Display Equipment
VGA_R[3]	A17	FPGA Module	VGA Port	Display Equipment
VGA_R[4]	G15	FPGA Module	VGA Port	Display Equipment
VGA_R[5]	G14	FPGA Module	VGA Port	Display Equipment
VGA_R[6]	A14	FPGA Module	VGA Port	Display Equipment
VGA_R[7]	B15	FPGA Module	VGA Port	Display Equipment
VGA_G[0]	G13	FPGA Module	VGA Port	Display Equipment
VGA_G[1]	F18	FPGA Module	VGA Port	Display Equipment
VGA_G[2]	B14	FPGA Module	VGA Port	Display Equipment
VGA_G[3]	A23	FPGA Module	VGA Port	Display Equipment
VGA_G[4]	F17	FPGA Module	VGA Port	Display Equipment
VGA_G[5]	G18	FPGA Module	VGA Port	Display Equipment
VGA_G[6]	A22	FPGA Module	VGA Port	Display Equipment
VGA_G[7]	A21	FPGA Module	VGA Port	Display Equipment
/VGA_BLANK	B16	FPGA Module	VGA Port	Display Equipment
/VGA_SYNC	C16	FPGA Module	VGA Port	Display Equipment
VGA_CLK	F23	FPGA Module	VGA Port	Display Equipment
VGA_SYNC_T	J24	FPGA Module	VGA Port	Display Equipment

VGA_HSYNC	J23	FPGA Module	VGA Port	Display Equipment
VGA_VSYNC	C15	FPGA Module	VGA Port	Display Equipment
VGA_M1	J26	FPGA Module	VGA Port	Display Equipment
VGA_M2	J25	FPGA Module	VGA Port	Display Equipment
22. Serial PORT				
Net Name	***PAD***	***From***	***To***	***비고***
UART_TXD1	F7	FPGA Module	Serial Port	UART control
UART_RXD1	H3	Serial Port	FPGA Module	UART control
23. PS2 PORT				
Net Name	***PAD***	***From***	***To***	***비고***
PS2_CLK	G6	PS2 Port	FPGA Module	PS2 control
PS2_DATA	G5	PS2 Port	FPGA Module	PS2 control
24. USB Serial PORT				
Net Name	***PAD***	***From***	***To***	***비고***
USB_TXD	H1	FPGA Module	Serial Port	UART control
USB_RXD	H2	Serial Port	FPGA Module	UART control
25. Expansion PORT - 1				
Net Name	***PAD***	***From***	***To***	***비고***
EXT1[1]	+5V	-	-	Power +5V
EXT1[2]	+5V	-	-	Power +5V
EXT1[3]	+12V	-	-	Power +12V
EXT1[4]	+12V	-	-	Power +12V
EXT1[5]	A20	FPGA Module	Base board	
EXT1[6]	A19	FPGA Module	Base board	
EXT1[7]	A18	FPGA Module	Base board	
EXT1[8]	A17	FPGA Module	Base board	
EXT1[9]	A14	FPGA Module	Base board	
EXT1[10]	B15	FPGA Module	Base board	
EXT1[11]	B14	FPGA Module	Base board	
EXT1[12]	A23	FPGA Module	Base board	
EXT1[13]	A22	FPGA Module	Base board	
EXT1[14]	A21	FPGA Module	Base board	
EXT1[15]	B20	FPGA Module	Base board	
EXT1[16]	B19	FPGA Module	Base board	
EXT1[17]	B18	FPGA Module	Base board	
EXT1[18]	B17	FPGA Module	Base board	
EXT1[19]	B16	FPGA Module	Base board	

EXT1[20]	C16	FPGA Module	Base board	
EXT1[21]	C15	FPGA Module	Base board	
EXT1[22]	B23	FPGA Module	Base board	
EXT1[23]	B22	FPGA Module	Base board	
EXT1[24]	B21	FPGA Module	Base board	
EXT1[25]	C23	FPGA Module	Base board	
EXT1[26]	C22	FPGA Module	Base board	
EXT1[27]	C21	FPGA Module	Base board	
EXT1[28]	C19	FPGA Module	Base board	
EXT1[29]	C17	FPGA Module	Base board	
EXT1[30]	D18	FPGA Module	Base board	
EXT1[31]	D17	FPGA Module	Base board	
EXT1[32]	D16	FPGA Module	Base board	
EXT1[33]	D15	FPGA Module	Base board	
EXT1[34]	D14	FPGA Module	Base board	
EXT1[35]	E18	FPGA Module	Base board	
EXT1[36]	E15	FPGA Module	Base board	
EXT1[37]	D21	FPGA Module	Base board	
EXT1[38]	D20	FPGA Module	Base board	
EXT1[39]	D19	FPGA Module	Base board	
EXT1[40]	F16	FPGA Module	Base board	
EXT1[41]	F15	FPGA Module	Base board	
EXT1[42]	F14	FPGA Module	Base board	
EXT1[43]	F13	FPGA Module	Base board	
EXT1[44]	E20	FPGA Module	Base board	
EXT1[45]	GND	-	-	Power GND
EXT1[46]	GND	-	-	Power GND
EXT1[47]	GND	-	-	Power GND
EXT1[48]	GND	-	-	Power GND
EXT1[49]	GND	-	-	Power GND
EXT1[50]	GND	-	-	Power GND
26. Expansion PORT - 2				
Net Name	***PAD***	***From***	***To***	***비고***
EXT2[1]	+5V	-	-	Power +5V
EXT2[2]	+5V	-	-	Power +5V
EXT2[3]	+12V	-	-	Power +12V
EXT2[4]	+12V	-	-	Power +12V
EXT2[5]	F17	FPGA Module	Base board	
EXT2[6]	G18	FPGA Module	Base board	

EXT2[7]	G17	FPGA Module	Base board	
EXT2[8]	G16	FPGA Module	Base board	
EXT2[9]	F23	FPGA Module	Base board	
EXT2[10]	F21	FPGA Module	Base board	
EXT2[11]	J22	FPGA Module	Base board	
EXT2[12]	K19	FPGA Module	Base board	
EXT2[13]	J26	FPGA Module	Base board	
EXT2[14]	J25	FPGA Module	Base board	
EXT2[15]	J24	FPGA Module	Base board	
EXT2[16]	J23	FPGA Module	Base board	
EXT2[17]	K25	FPGA Module	Base board	
EXT2[18]	K24	FPGA Module	Base board	
EXT2[19]	K23	FPGA Module	Base board	
EXT2[20]	K22	FPGA Module	Base board	
EXT2[21]	K21	FPGA Module	Base board	
EXT2[22]	L23	FPGA Module	Base board	
EXT2[23]	L21	FPGA Module	Base board	
EXT2[24]	L20	FPGA Module	Base board	
EXT2[25]	L19	FPGA Module	Base board	
EXT2[26]	K26	FPGA Module	Base board	
EXT2[27]	M21	FPGA Module	Base board	
EXT2[28]	M20	FPGA Module	Base board	
EXT2[29]	M19	FPGA Module	Base board	
EXT2[30]	L25	FPGA Module	Base board	
EXT2[31]	L24	FPGA Module	Base board	
EXT2[32]	N20	FPGA Module	Base board	
EXT2[33]	M25	FPGA Module	Base board	
EXT2[34]	M24	FPGA Module	Base board	
EXT2[35]	M23	FPGA Module	Base board	
EXT2[36]	M22	FPGA Module	Base board	
EXT2[37]	N24	FPGA Module	Base board	
EXT2[38]	N23	FPGA Module	Base board	
EXT2[39]	N21	FPGA Module	Base board	
EXT2[40]	V21	FPGA Module	Base board	
EXT2[41]	V20	FPGA Module	Base board	
EXT2[42]	AA23	FPGA Module	Base board	
EXT2[43]	Y26	FPGA Module	Base board	
EXT2[44]	Y25	FPGA Module	Base board	
EXT2[45]	GND	-	-	Power GND
EXT2[46]	GND	-	-	Power GND

EXT2[47]	GND	-	-	Power GND
EXT2[48]	GND	-	-	Power GND
EXT2[49]	GND	-	-	Power GND
EXT2[50]	GND	-	-	Power GND
27. AMBA BUS Interface				
Net Name	***PAD***	***From***	***To***	***비고***
H_ADDR[0]	A5	FPGA Module	Processor Module	
H_ADDR[1]	B8	FPGA Module	Processor Module	
H_ADDR[2]	B9	FPGA Module	Processor Module	
H_ADDR[3]	A10	FPGA Module	Processor Module	
H_ADDR[4]	AF21	FPGA Module	Processor Module	
H_ADDR[5]	B12	FPGA Module	Processor Module	
H_ADDR[6]	A4	FPGA Module	Processor Module	
H_ADDR[7]	AF22	FPGA Module	Processor Module	
H_ADDR[8]	A9	FPGA Module	Processor Module	
H_ADDR[9]	AF17	FPGA Module	Processor Module	
H_ADDR[10]	AE17	FPGA Module	Processor Module	
H_ADDR[11]	AF20	FPGA Module	Processor Module	
H_ADDR[12]	AE22	FPGA Module	Processor Module	
H_ADDR[13]	B7	FPGA Module	Processor Module	
H_ADDR[14]	AE18	FPGA Module	Processor Module	
H_ADDR[15]	AD21	FPGA Module	Processor Module	
H_ADDR[16]	AE23	FPGA Module	Processor Module	
H_ADDR[17]	AF19	FPGA Module	Processor Module	
H_ADDR[18]	AE21	FPGA Module	Processor Module	
H_ADDR[19]	AF18	FPGA Module	Processor Module	
H_ADDR[20]	AC22	FPGA Module	Processor Module	
H_ADDR[21]	AD22	FPGA Module	Processor Module	
H_ADDR[22]	AE19	FPGA Module	Processor Module	
H_ADDR[23]	AE16	FPGA Module	Processor Module	
H_ADDR[24]	AE20	FPGA Module	Processor Module	
H_ADDR[25]	AC17	FPGA Module	Processor Module	
H_ADDR[26]	AD15	FPGA Module	Processor Module	
H_ADDR[27]	AD19	FPGA Module	Processor Module	
H_ADDR[28]	AE15	FPGA Module	Processor Module	
H_ADDR[29]	AD23	FPGA Module	Processor Module	
H_ADDR[30]	AB20	FPGA Module	Processor Module	
H_ADDR[31]	AC21	FPGA Module	Processor Module	
H_DATA[0]	G10	FPGA Module	Processor Module	

H_DATA[1]	G12	FPGA Module	Processor Module	
H_DATA[2]	F2	FPGA Module	Processor Module	
H_DATA[3]	D10	FPGA Module	Processor Module	
H_DATA[4]	F11	FPGA Module	Processor Module	
H_DATA[5]	D2	FPGA Module	Processor Module	
H_DATA[6]	D9	FPGA Module	Processor Module	
H_DATA[7]	F9	FPGA Module	Processor Module	
H_DATA[8]	F10	FPGA Module	Processor Module	
H_DATA[9]	C12	FPGA Module	Processor Module	
H_DATA[10]	D5	FPGA Module	Processor Module	
H_DATA[11]	D12	FPGA Module	Processor Module	
H_DATA[12]	E8	FPGA Module	Processor Module	
H_DATA[13]	C8	FPGA Module	Processor Module	
H_DATA[14]	C7	FPGA Module	Processor Module	
H_DATA[15]	D11	FPGA Module	Processor Module	
H_DATA[16]	D8	FPGA Module	Processor Module	
H_DATA[17]	B11	FPGA Module	Processor Module	
H_DATA[18]	D7	FPGA Module	Processor Module	
H_DATA[19]	B5	FPGA Module	Processor Module	
H_DATA[20]	B10	FPGA Module	Processor Module	
H_DATA[21]	C10	FPGA Module	Processor Module	
H_DATA[22]	B6	FPGA Module	Processor Module	
H_DATA[23]	D6	FPGA Module	Processor Module	
H_DATA[24]	C11	FPGA Module	Processor Module	
H_DATA[25]	C9	FPGA Module	Processor Module	
H_DATA[26]	A7	FPGA Module	Processor Module	
H_DATA[27]	A8	FPGA Module	Processor Module	
H_DATA[28]	C4	FPGA Module	Processor Module	
H_DATA[29]	C6	FPGA Module	Processor Module	
H_DATA[30]	A6	FPGA Module	Processor Module	
H_DATA[31]	B4	FPGA Module	Processor Module	
H_RESET	F6	FPGA Module	Processor Module	
DMA_REQ0	G3	FPGA Module	Processor Module	
DMA_CLR0	G4	FPGA Module	Processor Module	
DMA_TC0	E5	FPGA Module	Processor Module	
DMA_REQ1	F4	FPGA Module	Processor Module	
DMA_CLR1	C3	FPGA Module	Processor Module	
DMA_TC1	F1	FPGA Module	Processor Module	
DMA_REQ2	E2	FPGA Module	Processor Module	
DMA_CLR2	G9	FPGA Module	Processor Module	

DMA_TC2	F3	FPGA Module	Processor Module	
DMA_REQ3	D1	FPGA Module	Processor Module	
DMA_CLR3	C2	FPGA Module	Processor Module	
DMA_TC3	F12	FPGA Module	Processor Module	
INT0	E1	FPGA Module	Processor Module	
INT1	G11	FPGA Module	Processor Module	
INT2	E12	FPGA Module	Processor Module	
INT3	E10	FPGA Module	Processor Module	
SPLIT0	Y16	FPGA Module	Processor Module	
SPLIT1	AB25	FPGA Module	Processor Module	
SPLIT2	AD24	FPGA Module	Processor Module	
SPLIT3	AB26	FPGA Module	Processor Module	
MASTER0	AA16	FPGA Module	Processor Module	
MASTER1	AC26	FPGA Module	Processor Module	
MASTER2	AA15	FPGA Module	Processor Module	
MASTER3	Y15	FPGA Module	Processor Module	
H_SEL	Y14	FPGA Module	Processor Module	
SLAVE_READYIN	AB18	FPGA Module	Processor Module	
SLAVE_READYOUT	AA14	FPGA Module	Processor Module	
MASTER_READY	AC25	FPGA Module	Processor Module	
GRANT	AA20	FPGA Module	Processor Module	
BUSREQ	Y13	FPGA Module	Processor Module	
MASTLOCK	AC23	FPGA Module	Processor Module	
LOCK	AC14	FPGA Module	Processor Module	
PROT0	AA18	FPGA Module	Processor Module	
PROT1	AB15	FPGA Module	Processor Module	
PROT2	AC24	FPGA Module	Processor Module	
PROT3	AA13	FPGA Module	Processor Module	
TRANS0	AD25	FPGA Module	Processor Module	
TRANS1	AC19	FPGA Module	Processor Module	
RESP0	Y18	FPGA Module	Processor Module	
RESP1	AC15	FPGA Module	Processor Module	
BURST0	AD16	FPGA Module	Processor Module	
BURST1	AC20	FPGA Module	Processor Module	
BURST2	AB21	FPGA Module	Processor Module	
WRITE	AA17	FPGA Module	Processor Module	
SIZE0	AD17	FPGA Module	Processor Module	
SIZE1	AC18	FPGA Module	Processor Module	
SIZE2	AC16	FPGA Module	Processor Module	
AHB_CLK	AF14	FPGA Module	Processor Module	

EASIC_PI_CLK	AA25	FPGA Module	Processor Module	

2) Xilinx Device Module

1. Clock				
Net Name	***PAD***	***From***	***To***	***비고***
CLK_MAIN[0]	F13	Oscillator	FPGA Device	MAIN CLOCK
CLK_MAIN[1]	AF14	Oscillator	FPGA Device	MAIN CLOCK
CLK_USER[0]	B14	Oscillator	FPGA Device	USER CLOCK
CLK_USER[1]	AF13	Oscillator	FPGA Device	USER CLOCK
2. RESET				
Net Name	***PAD***	***From***	***To***	***비고***
RESET	AB7	Button Switch	FPA Device	System Switch
3. LED				
Net Name	***PAD***	***From***	***To***	***비고***
LED_D[0]	Y2	FPGA Module	Base board	Display LED
LED_D[1]	Y1	FPGA Module	Base board	Display LED
LED_D[2]	AA10	FPGA Module	Base board	Display LED
LED_D[3]	AA7	FPGA Module	Base board	Display LED
LED_D[4]	AA3	FPGA Module	Base board	Display LED
LED_D[5]	AA2	FPGA Module	Base board	Display LED
LED_D[6]	Y10	FPGA Module	Base board	Display LED
LED_D[7]	AB12	FPGA Module	Base board	Display LED
LED_D[8]	AB9	FPGA Module	Base board	Display LED
LED_D[9]	AB1	FPGA Module	Base board	Display LED
LED_D[10]	Y20	FPGA Module	Base board	Display LED
LED_D[11]	AD6	FPGA Module	Base board	Display LED
LED_D[12]	J4	FPGA Module	Base board	Display LED
LED_D[13]	J5	FPGA Module	Base board	Display LED
LED_D[14]	Y23	FPGA Module	Base board	Display LED
LED_D[15]	AA12	FPGA Module	Base board	Display LED
4. Segment(Array)				
Net Name	***PAD***	***From***	***To***	***비고***
SEG_SCAN[0]	P9	FPGA Module	Base board	Display Segment
SEG_SCAN[1]	P8	FPGA Module	Base board	Display Segment

SEG_SCAN[2]	P7	FPGA Module	Base board	Display Segment
SEG_SCAN[3]	P6	FPGA Module	Base board	Display Segment
SEG_SCAN[4]	R4	FPGA Module	Base board	Display Segment
SEG_SCAN[5]	R3	FPGA Module	Base board	Display Segment
SEG_SCAN[6]	R2	FPGA Module	Base board	Display Segment
SEG_SCAN[7]	R1	FPGA Module	Base board	Display Segment
SEG_DATA_A	N4	FPGA Module	Base board	Display Segment
SEG_DATA_B	N2	FPGA Module	Base board	Display Segment
SEG_DATA_C	P4	FPGA Module	Base board	Display Segment
SEG_DATA_D	P3	FPGA Module	Base board	Display Segment
SEG_DATA_E	P2	FPGA Module	Base board	Display Segment
SEG_DATA_F	P1	FPGA Module	Base board	Display Segment
SEG_DATA_G	N9	FPGA Module	Base board	Display Segment
SEG_DATA_DP	P10	FPGA Module	Base board	Display Segment
5. Segment				
Net Name	***PAD***	***From***	***To***	***비고***
SEG_DATA_A	N1	FPGA Module	Base board	Display Segment
SEG_DATA_B	M10	FPGA Module	Base board	Display Segment
SEG_DATA_C	M9	FPGA Module	Base board	Display Segment
SEG_DATA_D	M8	FPGA Module	Base board	Display Segment
SEG_DATA_E	M7	FPGA Module	Base board	Display Segment
SEG_DATA_F	N7	FPGA Module	Base board	Display Segment
SEG_DATA_G	N6	FPGA Module	Base board	Display Segment
SEG_DATA_DP	N5	FPGA Module	Base board	Display Segment
6. Vacuum Fluorescent Display(VFD)				
Net Name	***PAD***	***From***	***To***	***비고***
VFD_D[0]	L10	FPGA Module	Base board	Display LCD
VFD_D[1]	L9	FPGA Module	Base board	Display LCD
VFD_D[2]	L7	FPGA Module	Base board	Display LCD
VFD_D[3]	L4	FPGA Module	Base board	Display LCD
VFD_D[4]	M6	FPGA Module	Base board	Display LCD
VFD_D[5]	M5	FPGA Module	Base board	Display LCD
VFD_D[6]	M4	FPGA Module	Base board	Display LCD
VFD_D[7]	M3	FPGA Module	Base board	Display LCD
VFD_E	M1	FPGA Module	Base board	Display LCD
VFD_RW	K8	FPGA Module	Base board	Display LCD
VFD_RS	K9	FPGA Module	Base board	Display LCD

7. DOT				
Net Name	***PAD***	***From***	***To***	***비고***
DOT_D[0]	N24	FPGA Module	Base board	Display Dot-matrix
DOT_D[1]	D16	FPGA Module	Base board	Display Dot-matrix
DOT_D[2]	C26	FPGA Module	Base board	Display Dot-matrix
DOT_D[3]	N21	FPGA Module	Base board	Display Dot-matrix
DOT_D[4]	N20	FPGA Module	Base board	Display Dot-matrix
DOT_D[5]	C25	FPGA Module	Base board	Display Dot-matrix
DOT_D[6]	D24	FPGA Module	Base board	Display Dot-matrix
DOT_D[7]	N19	FPGA Module	Base board	Display Dot-matrix
DOT_D[8]	N18	FPGA Module	Base board	Display Dot-matrix
DOT_D[9]	D23	FPGA Module	Base board	Display Dot-matrix
DOT_D[10]	D22	FPGA Module	Base board	Display Dot-matrix
DOT_D[11]	P23	FPGA Module	Base board	Display Dot-matrix
DOT_D[12]	P22	FPGA Module	Base board	Display Dot-matrix
DOT_D[13]	D21	FPGA Module	Base board	Display Dot-matrix
DOT_COM[0]	C21	FPGA Module	Base board	Display Dot-matrix
DOT_COM[1]	N17	FPGA Module	Base board	Display Dot-matrix
DOT_COM[2]	M26	FPGA Module	Base board	Display Dot-matrix
DOT_COM[3]	C20	FPGA Module	Base board	Display Dot-matrix
DOT_COM[4]	C18	FPGA Module	Base board	Display Dot-matrix
DOT_COM[5]	M25	FPGA Module	Base board	Display Dot-matrix
DOT_COM[6]	M23	FPGA Module	Base board	Display Dot-matrix
DOT_COM[7]	D18	FPGA Module	Base board	Display Dot-matrix
DOT_COM[8]	D17	FPGA Module	Base board	Display Dot-matrix
DOT_COM[9]	M22	FPGA Module	Base board	Display Dot-matrix
8. PIEZO				
Net Name	***PAD***	***From***	***To***	***비고***
PIEZO	M2	FPGA Module	Base board	Speaker control
9. SRAM				
Net Name	***PAD***	***From***	***To***	***비고***
SRAM_A[0]	G17	FPGA Module	Base board	Memory control
SRAM_A[1]	G19	FPGA Module	Base board	Memory control
SRAM_A[2]	G20	FPGA Module	Base board	Memory control
SRAM_A[3]	G21	FPGA Module	Base board	Memory control
SRAM_A[4]	G22	FPGA Module	Base board	Memory control
SRAM_A[5]	G23	FPGA Module	Base board	Memory control
SRAM_A[6]	G24	FPGA Module	Base board	Memory control

SRAM_A[7]	H15	FPGA Module	Base board	Memory control
SRAM_A[8]	H17	FPGA Module	Base board	Memory control
SRAM_A[9]	H20	FPGA Module	Base board	Memory control
SRAM_A[10]	H21	FPGA Module	Base board	Memory control
SRAM_A[11]	J16	FPGA Module	Base board	Memory control
SRAM_A[12]	J19	FPGA Module	Base board	Memory control
SRAM_A[13]	J22	FPGA Module	Base board	Memory control
SRAM_A[14]	J23	FPGA Module	Base board	Memory control
SRAM_A[15]	J25	FPGA Module	Base board	Memory control
SRAM_A[16]	J26	FPGA Module	Base board	Memory control
SRAM_A[17]	K16	FPGA Module	Base board	Memory control
SRAM_D[0]	R25	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[1]	R26	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[2]	T17	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[3]	T18	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[4]	T20	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[5]	T23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[6]	T24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[7]	U15	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[8]	U16	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[9]	U18	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[10]	U19	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[11]	U20	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[12]	U21	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[13]	U22	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[14]	U23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[15]	U24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[16]	V14	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[17]	V15	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[18]	V18	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[19]	V19	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[20]	V21	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[21]	V22	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[22]	V23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[23]	V24	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[24]	V25	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[25]	W15	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[26]	W17	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[27]	W20	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[28]	W21	FPGA & Base board	FPGA & Base board	Memory control

SRAM_D[29]	W23	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[30]	Y15	FPGA & Base board	FPGA & Base board	Memory control
SRAM_D[31]	Y17	FPGA & Base board	FPGA & Base board	Memory control
SRAM1_LB	K18	FPGA Module	Base board	Memory control
SRAM1_UB	K19	FPGA Module	Base board	Memory control
SRMA2_LB	K23	FPGA Module	Base board	Memory control
SRAM2_UB	K25	FPGA Module	Base board	Memory control
SRAM_OE	L20	FPGA Module	Base board	Memory control
SRAM_WE	L18	FPGA Module	Base board	Memory control
SRAM_CS	L17	FPGA Module	Base board	Memory control
10. Flash Memory				
Net Name	***PAD***	***From***	***To***	***비고***
Flash_A[1]	G17	FPGA Module	Base board	Memory control
Flash_A[2]	G19	FPGA Module	Base board	Memory control
Flash_A[3]	G20	FPGA Module	Base board	Memory control
Flash_A[4]	G21	FPGA Module	Base board	Memory control
Flash_A[5]	G22	FPGA Module	Base board	Memory control
Flash_A[6]	G23	FPGA Module	Base board	Memory control
Flash_A[7]	G24	FPGA Module	Base board	Memory control
Flash_A[8]	H15	FPGA Module	Base board	Memory control
Flash_A[9]	H17	FPGA Module	Base board	Memory control
Flash_A[10]	H20	FPGA Module	Base board	Memory control
Flash_A[11]	H21	FPGA Module	Base board	Memory control
Flash_A[12]	J16	FPGA Module	Base board	Memory control
Flash_A[13]	J19	FPGA Module	Base board	Memory control
Flash_A[14]	J22	FPGA Module	Base board	Memory control
Flash_A[15]	J23	FPGA Module	Base board	Memory control
Flash_A[16]	J25	FPGA Module	Base board	Memory control
Flash_A[17]	J26	FPGA Module	Base board	Memory control
Flash_A[18]	K16	FPGA Module	Base board	Memory control
Flash_A[19]	K18	FPGA Module	Base board	Memory control
Flash_A[20]	K19	FPGA Module	Base board	Memory control
Flash_DQ[0]	R25	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[1]	R26	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[2]	T17	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[3]	T18	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[4]	T20	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[5]	T23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[6]	T24	FPGA & Base board	FPGA & Base board	Memory control

Flash_DQ[7]	U15	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[8]	U16	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[9]	U18	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[10]	U19	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[11]	U20	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[12]	U21	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[13]	U22	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[14]	U23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[15]	U24	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[16]	V14	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[17]	V15	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[18]	V18	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[19]	V19	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[20]	V21	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[21]	V22	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[22]	V23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[23]	V24	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[24]	V25	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[25]	W15	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[26]	W17	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[27]	W20	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[28]	W21	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[29]	W23	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[30]	Y15	FPGA & Base board	FPGA & Base board	Memory control
Flash_DQ[31]	Y17	FPGA & Base board	FPGA & Base board	Memory control
/Flash_OE	K20	FPGA Module	Base board	Memory control
/Flash_WE	K21	FPGA Module	Base board	Memory control
/Flash_CS	K22	FPGA Module	Base board	Memory control
/Flash_RP	K23	FPGA Module	Base board	Memory control
/Flash_RB	K25	FPGA Module	Base board	Memory control
11. Step Motor				
Net Name	***PAD***	***From***	***To***	***H/L***
STEP_DATA[0]	Y22	FPGA Module	Base board	Motor control (A)
STEP_DATA[1]	Y25	FPGA Module	Base board	Motor control (B)
STEP_DATA[2]	Y21	FPGA Module	Base board	Motor control (/A)
STEP_DATA[3]	Y24	FPGA Module	Base board	Motor control (/B)
MOTOR_SENSOR	Y26	Base board	FPGA Module	Magnetic sensor

12. SD Connector				
Net Name	***PAD***	***From***	***To***	***비고***
SD_DATA[0]	J1	FPGA & Base board	FPGA & Base board	Memory control
SD_DATA[1]	H6	FPGA & Base board	FPGA & Base board	Memory control
SD_DATA[2]	H4	FPGA & Base board	FPGA & Base board	Memory control
SD_DATA[3]	J8	FPGA & Base board	FPGA & Base board	Memory control
SD_CMD	J3	FPGA & Base board	FPGA & Base board	Memory control
SD_CLK	J2	FPGA & Base board	FPGA & Base board	Memory control
13. DIP SWITCH				
Net Name	***PAD***	***From***	***To***	***비고***
DIP_D[0]	A13	Base board	FPGA Module	Input Equipment
DIP_D[1]	C1	Base board	FPGA Module	Input Equipment
DIP_D[2]	C2	Base board	FPGA Module	Input Equipment
DIP_D[3]	C4	Base board	FPGA Module	Input Equipment
DIP_D[4]	A5	Base board	FPGA Module	Input Equipment
DIP_D[5]	A7	Base board	FPGA Module	Input Equipment
DIP_D[6]	D1	Base board	FPGA Module	Input Equipment
DIP_D[7]	D2	Base board	FPGA Module	Input Equipment
14. BUTTON SWITCH				
Net Name	***PAD***	***From***	***To***	***비고***
BUTTON_SW[0]	Y2	Base board	FPGA Module	Input Equipment
BUTTON_SW[1]	Y1	Base board	FPGA Module	Input Equipment
BUTTON_SW[2]	AA10	Base board	FPGA Module	Input Equipment
BUTTON_SW[3]	AA7	Base board	FPGA Module	Input Equipment
BUTTON_SW[4]	AA3	Base board	FPGA Module	Input Equipment
BUTTON_SW[5]	AA2	Base board	FPGA Module	Input Equipment
BUTTON_SW[6]	Y10	Base board	FPGA Module	Input Equipment
BUTTON_SW[7]	AB12	Base board	FPGA Module	Input Equipment
BUTTON_SW[8]	AB9	Base board	FPGA Module	Input Equipment
BUTTON_SW[9]	AB1	Base board	FPGA Module	Input Equipment
BUTTON_SW[10]	Y20	Base board	FPGA Module	Input Equipment
BUTTON_SW[11]	AD6	Base board	FPGA Module	Input Equipment
BUTTON_SW[12]	J4	Base board	FPGA Module	Input Equipment
BUTTON_SW[13]	J5	Base board	FPGA Module	Input Equipment
BUTTON_SW[14]	Y23	Base board	FPGA Module	Input Equipment
BUTTON_SW[15]	AA12	Base board	FPGA Module	Input Equipment
15. Humidity & Temp Sensor				

Net Name	PAD	From	To	비고
SENSOR_CLK	K12	FPGA Module	Base board	Serial clock input
SENSOR_DATA	K11	FPGA & Base board	FPGA & Base board	Serial data
16. TFT-LCD				
Net Name	**PAD**	**From**	**To**	**비고**
TFTLCD_D[0]	V12	FPGA Module	TFTLCD Module	TFTLCD RED[0]
TFTLCD_D[1]	V11	FPGA Module	TFTLCD Module	TFTLCD RED[1]
TFTLCD_D[2]	V7	FPGA Module	TFTLCD Module	TFTLCD RED[2]
TFTLCD_D[3]	V6	FPGA Module	TFTLCD Module	TFTLCD RED[3]
TFTLCD_D[4]	V5	FPGA Module	TFTLCD Module	TFTLCD RED[4]
TFTLCD_D[5]	V2	FPGA Module	TFTLCD Module	TFTLCD RED[5]
TFTLCD_D[6]	W3	FPGA Module	TFTLCD Module	TFTLCD RED[6]
TFTLCD_D[7]	V13	FPGA Module	TFTLCD Module	TFTLCD RED[7]
TFTLCD_D[8]	V10	FPGA Module	TFTLCD Module	TFTLCD GREEN[0]
TFTLCD_D[9]	W7	FPGA Module	TFTLCD Module	TFTLCD GREEN[1]
TFTLCD_D[10]	W6	FPGA Module	TFTLCD Module	TFTLCD GREEN[2]
TFTLCD_D[11]	W4	FPGA Module	TFTLCD Module	TFTLCD GREEN[3]
TFTLCD_D[12]	V4	FPGA Module	TFTLCD Module	TFTLCD GREEN[4]
TFTLCD_D[13]	U3	FPGA Module	TFTLCD Module	TFTLCD GREEN[5]
TFTLCD_D[14]	Y8	FPGA Module	TFTLCD Module	TFTLCD GREEN[6]
TFTLCD_D[15]	Y4	FPGA Module	TFTLCD Module	TFTLCD GREEN[7]
TFTLCD_D[16]	Y3	FPGA Module	TFTLCD Module	TFTLCD_BLUE[0]
TFTLCD_D[17]	W2	FPGA Module	TFTLCD Module	TFTLCD_BLUE[1]
TFTLCD_D[18]	W1	FPGA Module	TFTLCD Module	TFTLCD_BLUE[2]
TFTLCD_D[19]	Y11	FPGA Module	TFTLCD Module	TFTLCD_BLUE[3]
TFTLCD_D[20]	W13	FPGA Module	TFTLCD Module	TFTLCD_BLUE[4]
TFTLCD_D[21]	W12	FPGA Module	TFTLCD Module	TFTLCD_BLUE[5]
TFTLCD_D[22]	W10	FPGA Module	TFTLCD Module	TFTLCD_BLUE[6]
TFTLCD_D[23]	W9	FPGA Module	TFTLCD Module	TFTLCD_BLUE[7]
TFTLCD_ENABLE	Y9	FPGA Module	TFTLCD Module	Enable signal
TFTLCD_LCLK	Y6	FPGA Module	TFTLCD Module	HSYNC
TFTLCD_FCLK	Y5	FPGA Module	TFTLCD Module	VSYNC
TFTLCD_PCLK	R8	FPGA Module	TFTLCD Module	CLOCK
TFTLCD_RESET	U9	FPGA Module	TFTLCD Module	RESET
17. Image Sensor				
Net Name	**PAD**	**From**	**To**	**비고**
CIS DATA[]	T9	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	T7	FPGA Module	TFTLCD Module	Image out Data

CIS DATA[]	T5	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	T4	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	U6	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	U5	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	U4	FPGA Module	TFTLCD Module	Image out Data
CIS DATA[]	U2	FPGA Module	TFTLCD Module	Image out Data
CIS_SCK	U1	FPGA Module	TFTLCD Module	Serial clock
CIS_SDA	V1	FPGA Module	TFTLCD Module	Serial data
CIS_STROBE	U11	FPGA Module	TFTLCD Module	Strobe signal
CIS_VCLK	U8	FPGA Module	TFTLCD Module	Pixel clock
CIS_LVALID	U7	FPGA Module	TFTLCD Module	Line valid
CIS_FSYNC	V8	FPGA Module	TFTLCD Module	Frame valid
CIS_MCLK	R9	FPGA Module	TFTLCD Module	Master clock
CIS_RESET	U9	FPGA Module	TFTLCD Module	reset

18. TFT-LCD Touch

Net Name	*PAD*	*From*	*To*	*비고*
TOUCH_DCLK	T10	FPGA Module	TFTLCD Module	External clock
TOUCH_DIN	R6	FPGA Module	TFTLCD Module	Data input
TOUCH_DOUT	R10	FPGA Module	TFTLCD Module	Data output
TOUCH_BUSY	R5	FPGA Module	TFTLCD Module	Busy output
TOUCH_IRQ	R7	FPGA Module	TFTLCD Module	Pen interrupt
TOUCH_CS	T3	FPGA Module	TFTLCD Module	Chip Select

19. OLED

Net Name	*PAD*	*From*	*To*	*비고*
OLED_DATA[0]	P18	FPGA Module	OLED Module	OLED Data
OLED_DATA[1]	R19	FPGA Module	OLED Module	OLED Data
OLED_DATA[2]	E14	FPGA Module	OLED Module	OLED Data
OLED_DATA[3]	D26	FPGA Module	OLED Module	OLED Data
OLED_DATA[4]	R18	FPGA Module	OLED Module	OLED Data
OLED_DATA[5]	R17	FPGA Module	OLED Module	OLED Data
OLED_DATA[6]	D25	FPGA Module	OLED Module	OLED Data
OLED_DATA[7]	F15	FPGA Module	OLED Module	OLED Data
OLED_CS	D20	FPGA Module	OLED Module	Chip select
OLED_WN	E17	FPGA Module	OLED Module	Write enable
OLED_RN	E15	FPGA Module	OLED Module	Read enable
OLED_DC	P20	FPGA Module	OLED Module	Data/Command
OLED_RESET	P21	FPGA Module	OLED Module	Reset pin

20. Audio Codec				
Net Name	***PAD***	***From***	***To***	***비고***
AIC_CS	P26	FPGA Module	Audio Codec	Latch/Add Select
AIC_MODE	P25	FPGA Module	Audio Codec	Serial interface
AIC_SDI	F14	FPGA Module	Audio Codec	Serial Data input
AIC_SCLK	E26	FPGA Module	Audio Codec	Serial Data clock
AIC_MCLK	J20	FPGA Module	Audio Codec	External clock
AIC_LRCOUT	R21	FPGA Module	Audio Codec	ADC clock signal
AIC_DOUT	E24	FPGA Module	Audio Codec	Serial Data output
AIC_LRCIN	E21	FPGA Module	Audio Codec	DAC clock signal
AIC_DIN	R22	FPGA Module	Audio Codec	Serial Data input
AIC_BCLK	V17	FPGA Module	Audio Codec	Serial bit clock
21. VGA PORT				
Net Name	***PAD***	***From***	***To***	***비고***
VGA_B[0]	F25	FPGA Module	VGA Port	Display Equipment
VGA_B[1]	F24	FPGA Module	VGA Port	Display Equipment
VGA_B[2]	C17	FPGA Module	VGA Port	Display Equipment
VGA_B[3]	C16	FPGA Module	VGA Port	Display Equipment
VGA_B[4]	L22	FPGA Module	VGA Port	Display Equipment
VGA_B[5]	M21	FPGA Module	VGA Port	Display Equipment
VGA_B[6]	C15	FPGA Module	VGA Port	Display Equipment
VGA_B[7]	B23	FPGA Module	VGA Port	Display Equipment
VGA_R[0]	A22	FPGA Module	VGA Port	Display Equipment
VGA_R[1]	A20	FPGA Module	VGA Port	Display Equipment
VGA_R[2]	A19	FPGA Module	VGA Port	Display Equipment
VGA_R[3]	A18	FPGA Module	VGA Port	Display Equipment
VGA_R[4]	F23	FPGA Module	VGA Port	Display Equipment
VGA_R[5]	F22	FPGA Module	VGA Port	Display Equipment
VGA_R[6]	A15	FPGA Module	VGA Port	Display Equipment
VGA_R[7]	B20	FPGA Module	VGA Port	Display Equipment
VGA_G[0]	F20	FPGA Module	VGA Port	Display Equipment
VGA_G[1]	F19	FPGA Module	VGA Port	Display Equipment
VGA_G[2]	B19	FPGA Module	VGA Port	Display Equipment
VGA_G[3]	B18	FPGA Module	VGA Port	Display Equipment
VGA_G[4]	F17	FPGA Module	VGA Port	Display Equipment
VGA_G[5]	G15	FPGA Module	VGA Port	Display Equipment
VGA_G[6]	B17	FPGA Module	VGA Port	Display Equipment
VGA_G[7]	B15	FPGA Module	VGA Port	Display Equipment
/VGA_BLANK	B21	FPGA Module	VGA Port	Display Equipment

/VGA_SYNC	C23	FPGA Module	VGA Port	Display Equipment
VGA_CLK	J21	FPGA Module	VGA Port	Display Equipment
VGA_SYNC_T	M18	FPGA Module	VGA Port	Display Equipment
VGA_HSYNC	L24	FPGA Module	VGA Port	Display Equipment
VGA_VSYNC	C22	FPGA Module	VGA Port	Display Equipment
VGA_M1	M20	FPGA Module	VGA Port	Display Equipment
VGA_M2	M19	FPGA Module	VGA Port	Display Equipment
22. Serial PORT				
Net Name	***PAD***	***From***	***To***	***비고***
UART_TXD1	H12	FPGA Module	Serial Port	UART control
UART_RXD1	K2	Serial Port	FPGA Module	UART control
23. PS2 PORT				
Net Name	***PAD***	***From***	***To***	***비고***
PS2_CLK	J9	PS2 Port	FPGA Module	PS2 control
PS2_DATA	J8	PS2 Port	FPGA Module	PS2 control
24. USB Serial PORT				
Net Name	***PAD***	***From***	***To***	***비고***
USB_TXD	J11	FPGA Module	Serial Port	UART control
USB_RXD	J12	Serial Port	FPGA Module	UART control
25. Expansion PORT - 1				
Net Name	***PAD***	***From***	***To***	***비고***
EXT1[1]	+5V	-	-	Power +5V
EXT1[2]	+5V	-	-	Power +5V
EXT1[3]	+12V	-	-	Power +12V
EXT1[4]	+12V	-	-	Power +12V
EXT1[5]	A22	FPGA Module	Base board	
EXT1[6]	A20	FPGA Module	Base board	
EXT1[7]	A19	FPGA Module	Base board	
EXT1[8]	A18	FPGA Module	Base board	
EXT1[9]	A15	FPGA Module	Base board	
EXT1[10]	B20	FPGA Module	Base board	
EXT1[11]	B19	FPGA Module	Base board	
EXT1[12]	B18	FPGA Module	Base board	
EXT1[13]	B17	FPGA Module	Base board	
EXT1[14]	B15	FPGA Module	Base board	
EXT1[15]	C17	FPGA Module	Base board	
EXT1[16]	C16	FPGA Module	Base board	

EXT1[17]	C15	FPGA Module	Base board	
EXT1[18]	B23	FPGA Module	Base board	
EXT1[19]	B21	FPGA Module	Base board	
EXT1[20]	C23	FPGA Module	Base board	
EXT1[21]	C22	FPGA Module	Base board	
EXT1[22]	C21	FPGA Module	Base board	
EXT1[23]	C20	FPGA Module	Base board	
EXT1[24]	C18	FPGA Module	Base board	
EXT1[25]	D18	FPGA Module	Base board	
EXT1[26]	D17	FPGA Module	Base board	
EXT1[27]	D16	FPGA Module	Base board	
EXT1[28]	C26	FPGA Module	Base board	
EXT1[29]	C25	FPGA Module	Base board	
EXT1[30]	D24	FPGA Module	Base board	
EXT1[31]	D23	FPGA Module	Base board	
EXT1[32]	D22	FPGA Module	Base board	
EXT1[33]	D21	FPGA Module	Base board	
EXT1[34]	D20	FPGA Module	Base board	
EXT1[35]	E17	FPGA Module	Base board	
EXT1[36]	E15	FPGA Module	Base board	
EXT1[37]	E14	FPGA Module	Base board	
EXT1[38]	D26	FPGA Module	Base board	
EXT1[39]	D25	FPGA Module	Base board	
EXT1[40]	F15	FPGA Module	Base board	
EXT1[41]	F14	FPGA Module	Base board	
EXT1[42]	E26	FPGA Module	Base board	
EXT1[43]	E24	FPGA Module	Base board	
EXT1[44]	E21	FPGA Module	Base board	
EXT1[45]	GND	-	-	Power GND
EXT1[46]	GND	-	-	Power GND
EXT1[47]	GND	-	-	Power GND
EXT1[48]	GND	-	-	Power GND
EXT1[49]	GND	-	-	Power GND
EXT1[50]	GND	-	-	Power GND

26. Expansion PORT - 2

Net Name	*PAD*	*From*	*To*	*비고*
EXT2[1]	+5V	-	-	Power +5V
EXT2[2]	+5V	-	-	Power +5V
EXT2[3]	+12V	-	-	Power +12V

EXT2[4]	+12V	-	-	Power +12V
EXT2[5]	F23	FPGA Module	Base board	
EXT2[6]	F22	FPGA Module	Base board	
EXT2[7]	F20	FPGA Module	Base board	
EXT2[8]	F19	FPGA Module	Base board	
EXT2[9]	F17	FPGA Module	Base board	
EXT2[10]	G15	FPGA Module	Base board	
EXT2[11]	F25	FPGA Module	Base board	
EXT2[12]	F24	FPGA Module	Base board	
EXT2[13]	J21	FPGA Module	Base board	
EXT2[14]	J20	FPGA Module	Base board	
EXT2[15]	L22	FPGA Module	Base board	
EXT2[16]	M21	FPGA Module	Base board	
EXT2[17]	M20	FPGA Module	Base board	
EXT2[18]	M19	FPGA Module	Base board	
EXT2[19]	M18	FPGA Module	Base board	
EXT2[20]	L24	FPGA Module	Base board	
EXT2[21]	N17	FPGA Module	Base board	
EXT2[22]	M26	FPGA Module	Base board	
EXT2[23]	M25	FPGA Module	Base board	
EXT2[24]	M23	FPGA Module	Base board	
EXT2[25]	M22	FPGA Module	Base board	
EXT2[26]	N24	FPGA Module	Base board	
EXT2[27]	N21	FPGA Module	Base board	
EXT2[28]	N20	FPGA Module	Base board	
EXT2[29]	N19	FPGA Module	Base board	
EXT2[30]	N18	FPGA Module	Base board	
EXT2[31]	P23	FPGA Module	Base board	
EXT2[32]	P22	FPGA Module	Base board	
EXT2[33]	P21	FPGA Module	Base board	
EXT2[34]	P20	FPGA Module	Base board	
EXT2[35]	P18	FPGA Module	Base board	
EXT2[36]	R19	FPGA Module	Base board	
EXT2[37]	R18	FPGA Module	Base board	
EXT2[38]	R17	FPGA Module	Base board	
EXT2[39]	P26	FPGA Module	Base board	
EXT2[40]	P25	FPGA Module	Base board	
EXT2[41]	R22	FPGA Module	Base board	
EXT2[42]	R21	FPGA Module	Base board	
EXT2[43]	R20	FPGA Module	Base board	

EXT2[44]	V17	FPGA Module	Base board	
EXT2[45]	GND	-	-	Power GND
EXT2[46]	GND	-	-	Power GND
EXT2[47]	GND	-	-	Power GND
EXT2[48]	GND	-	-	Power GND
EXT2[49]	GND	-	-	Power GND
EXT2[50]	GND	-	-	Power GND
27. AMBA BUS Interface				
Net Name	***PAD***	***From***	***To***	***비고***
H_ADDR[0]	A4	FPGA Module	Processor Module	
H_ADDR[1]	B7	FPGA Module	Processor Module	
H_ADDR[2]	B8	FPGA Module	Processor Module	
H_ADDR[3]	B1	FPGA Module	Processor Module	
H_ADDR[4]	AD1	FPGA Module	Processor Module	
H_ADDR[5]	B12	FPGA Module	Processor Module	
H_ADDR[6]	A3	FPGA Module	Processor Module	
H_ADDR[7]	AD2	FPGA Module	Processor Module	
H_ADDR[8]	A12	FPGA Module	Processor Module	
H_ADDR[9]	AC8	FPGA Module	Processor Module	
H_ADDR[10]	AF20	FPGA Module	Processor Module	
H_ADDR[11]	AC12	FPGA Module	Processor Module	
H_ADDR[12]	AC3	FPGA Module	Processor Module	
H_ADDR[13]	B6	FPGA Module	Processor Module	
H_ADDR[14]	AF23	FPGA Module	Processor Module	
H_ADDR[15]	AE23	FPGA Module	Processor Module	
H_ADDR[16]	AC6	FPGA Module	Processor Module	
H_ADDR[17]	AC11	FPGA Module	Processor Module	
H_ADDR[18]	AC2	FPGA Module	Processor Module	
H_ADDR[19]	AC9	FPGA Module	Processor Module	
H_ADDR[20]	AE17	FPGA Module	Processor Module	
H_ADDR[21]	AE25	FPGA Module	Processor Module	
H_ADDR[22]	AF25	FPGA Module	Processor Module	
H_ADDR[23]	AF19	FPGA Module	Processor Module	
H_ADDR[24]	AC1	FPGA Module	Processor Module	
H_ADDR[25]	AD20	FPGA Module	Processor Module	
H_ADDR[26]	AE18	FPGA Module	Processor Module	
H_ADDR[27]	AE21	FPGA Module	Processor Module	
H_ADDR[28]	AF18	FPGA Module	Processor Module	
H_ADDR[29]	AE26	FPGA Module	Processor Module	

H_ADDR[30]	AC26	FPGA Module	Processor Module	
H_ADDR[31]	AD26	FPGA Module	Processor Module	
H_DATA[0]	F7	FPGA Module	Processor Module	
H_DATA[1]	F12	FPGA Module	Processor Module	
H_DATA[2]	H6	FPGA Module	Processor Module	
H_DATA[3]	D13	FPGA Module	Processor Module	
H_DATA[4]	F3	FPGA Module	Processor Module	
H_DATA[5]	G9	FPGA Module	Processor Module	
H_DATA[6]	D11	FPGA Module	Processor Module	
H_DATA[7]	E12	FPGA Module	Processor Module	
H_DATA[8]	F2	FPGA Module	Processor Module	
H_DATA[9]	D3	FPGA Module	Processor Module	
H_DATA[10]	D6	FPGA Module	Processor Module	
H_DATA[11]	E3	FPGA Module	Processor Module	
H_DATA[12]	E4	FPGA Module	Processor Module	
H_DATA[13]	C8	FPGA Module	Processor Module	
H_DATA[14]	C7	FPGA Module	Processor Module	
H_DATA[15]	E1	FPGA Module	Processor Module	
H_DATA[16]	D10	FPGA Module	Processor Module	
H_DATA[17]	B10	FPGA Module	Processor Module	
H_DATA[18]	D9	FPGA Module	Processor Module	
H_DATA[19]	B3	FPGA Module	Processor Module	
H_DATA[20]	B9	FPGA Module	Processor Module	
H_DATA[21]	C11	FPGA Module	Processor Module	
H_DATA[22]	B4	FPGA Module	Processor Module	
H_DATA[23]	D8	FPGA Module	Processor Module	
H_DATA[24]	C12	FPGA Module	Processor Module	
H_DATA[25]	C10	FPGA Module	Processor Module	
H_DATA[26]	A9	FPGA Module	Processor Module	
H_DATA[27]	A10	FPGA Module	Processor Module	
H_DATA[28]	C5	FPGA Module	Processor Module	
H_DATA[29]	C6	FPGA Module	Processor Module	
H_DATA[30]	B2	FPGA Module	Processor Module	
H_DATA[31]	A8	FPGA Module	Processor Module	
H_RESET	H10	FPGA Module	Processor Module	
DMA_REQ0	J6	FPGA Module	Processor Module	
DMA_CLR0	J7	FPGA Module	Processor Module	
DMA_TC0	H1	FPGA Module	Processor Module	
DMA_REQ1	H9	FPGA Module	Processor Module	
DMA_CLR1	G4	FPGA Module	Processor Module	

DMA_TC1	H2	FPGA Module	Processor Module	
DMA_REQ2	G12	FPGA Module	Processor Module	
DMA_CLR2	F5	FPGA Module	Processor Module	
DMA_TC2	H7	FPGA Module	Processor Module	
DMA_REQ3	G6	FPGA Module	Processor Module	
DMA_CLR3	G3	FPGA Module	Processor Module	
DMA_TC3	F4	FPGA Module	Processor Module	
INT0	G10	FPGA Module	Processor Module	
INT1	F8	FPGA Module	Processor Module	
INT2	E10	FPGA Module	Processor Module	
INT3	E7	FPGA Module	Processor Module	
SPLIT0	AB26	FPGA Module	Processor Module	
SPLIT1	AA17	FPGA Module	Processor Module	
SPLIT2	AB15	FPGA Module	Processor Module	
SPLIT3	AA18	FPGA Module	Processor Module	
MASTER0	AC20	FPGA Module	Processor Module	
MASTER1	AA25	FPGA Module	Processor Module	
MASTER2	AC19	FPGA Module	Processor Module	
MASTER3	AB24	FPGA Module	Processor Module	
H_SEL	AB23	FPGA Module	Processor Module	
SLAVE_READYIN	AC25	FPGA Module	Processor Module	
SLAVE_READYOUT	AC16	FPGA Module	Processor Module	
MASTER_READY	AA24	FPGA Module	Processor Module	
GRANT	AC23	FPGA Module	Processor Module	
BUSREQ	AB18	FPGA Module	Processor Module	
MASTLOCK	AA22	FPGA Module	Processor Module	
LOCK	AD15	FPGA Module	Processor Module	
PROT0	AC22	FPGA Module	Processor Module	
PROT1	AC24	FPGA Module	Processor Module	
PROT2	AA23	FPGA Module	Processor Module	
PROT3	AC15	FPGA Module	Processor Module	
TRANS0	AB16	FPGA Module	Processor Module	
TRANS1	AD22	FPGA Module	Processor Module	
RESP0	AC14	FPGA Module	Processor Module	
RESP1	AD17	FPGA Module	Processor Module	
BURST0	AE19	FPGA Module	Processor Module	
BURST1	AD25	FPGA Module	Processor Module	
BURST2	AD14	FPGA Module	Processor Module	
WRITE	AC21	FPGA Module	Processor Module	
SIZE0	AE20	FPGA Module	Processor Module	

SIZE1	AD21	FPGA Module	Processor Module	
SIZE2	AD19	FPGA Module	Processor Module	
AHB_CLK		FPGA Module	Processor Module	
EASIC_PI_CLK		FPGA Module	Processor Module	